C4-3a: Intro to Relational Database

Database Concept

Why use a Relational Database?

Customers



Products



Orders



https://www.youtube.com/watch?v=znw_rnnnWjY

Data Integrity

Data integrity refers to the accuracy, consistency, and reliability of data stored in a database. It ensures that the data remains correct and unaltered during various operations such as storage, retrieval, and processing.

Accuracy: Data integrity ensures that the data entered into the database is accurate and free from errors. Validation rules are applied during data entry to ensure that the data is reasonable and within expected parameters.

Consistency: This means that data remains consistent across the database. If the same data appears in multiple places, all instances should be updated simultaneously. Constraint rules, like PK and FK, are enforced to ensure data integrity across various tables.

Data Redundancy

Redundancy in database is having several copies of the same data within or across several tables.

It not only increases storage space and cost, but it may also cause inconsistency in data when they are not updated properly.

There are three types of anomalies: update, deletion, and insertion anomalies.

MatricNo	Name	Gender	CivicsClass	CivicsTutor
1	Alex	M	22S01	Peter Tan
2	Brandon	M	22S01	Peter Tan
3	Charles	M	22S01	Peter Tan
4	Dave	M	22S01	Peter Tan
5	Eliza	F	22S05	Jane Wong
6	Felicia	F	22S11	Mary Ng
7	George	M	22S11	Mary Ng
8	Hongli	M	22S11	Mary Ng

Data Redundancy – Update Anomaly

An update anomaly is a data inconsistency that results from data redundancy and a partial update.

- If the user performing the update does not realize that the same data is stored elsewhere, then the update will not be done properly.

If the Civics Tutor is changed to Mr John Lim and the table is not updated properly, then there will an update anomaly.

MatricNo	Name	Gender	CivicsClass	CivicsTutor
1	Alex	M	22S01	John Lim
2	Brandon	M	22S01	Peter Tan
3	Charles	M	22S01	Peter Tan
4	Dave	M	22S01	Peter Tan
5	Eliza	F	22S05	Jane Wong
6	Felicia	F	22S11	Mary Ng
7	George	M	22S11	Mary Ng
8	Hongli	M	22S11	Mary Ng

Data Redundancy – Deletion Anomaly

A deletion anomaly is the unintended loss of data due to deletion of other data. This results in database inconsistencies and is an example of how combining information that does not really belong together into one table can cause problems.

If the only student in 22S05 Eliza is deleted from this table, then the info about the Civics Tutor for 22S05 would be lost.

MatricNo	Name	Gender	CivicsClass	CivicsTutor
1	Alex	M	22S01	Peter Tan
2	Brandon	M	22S01	Peter Tan
3	Charles	M	22S01	Peter Tan
4	Dave	M	22S01	Peter Tan
5	Eliza	F	22S05	Jane Wong
6	Felicia	F	22S11	Mary Ng
7	George	M	22S11	Mary Ng
8	Hongli	M	22S11	Mary Ng

Data Redundancy – Insertion Anomaly

An insertion anomaly is the inability to add data to the database due to the absence of other data. This results in database inconsistencies due to omission.

If a new student Yan Lin is to be added into this table but she has not yet been assigned to a class, hence the info for CivicsClass and CivicsTutor will be omitted.

MatricNo	Name	Gender	CivicsClass	CivicsTutor
1	Alex	M	22S01	Peter Tan
2	Brandon	M	22S01	Peter Tan
3	Charles	M	22S01	Peter Tan
4	Dave	M	22S01	Peter Tan
5	Eliza	F	22S05	Jane Wong
6	Felicia	F	22S11	Mary Ng
7	George	M	22S11	Mary Ng
8	Hongli	M	22S11	Mary Ng
9	Yan Lin	F		

Data Dependencies

Redundancy in relational database can be avoided by **database normalisation**.

But before we could learn about normalisation, we need to first understand about data dependencies, specifically **functional** and **transitive dependencies**.

Dependency is the relationship between attributes of a table related to each other.

Functional Dependency

If attribute Y is functionally dependent on attribute X, the value of X uniquely determines the value of Y:

$$X \rightarrow Y$$

MatricNo uniquely identifies the Name because if we know the MatricNo, we can know the Name associated with it. Therefore, we can say Name is functionally dependent on MatricNo:

(MatricNo → Name)

MatricNo	Name	Gender	CivicsClass	CivicsTutor
1	Alex	M	22S01	Peter Tan
2	Brandon	M	22S01	Peter Tan
3	Charles	M	22S01	Peter Tan
4	Dave	M	22S01	Peter Tan
5	Eliza	F	22S05	Jane Wong
6	Felicia	F	22S11	Mary Ng
7	George	M	22S11	Mary Ng
8	Hongli	M	22S11	Mary Ng

Transitive Dependency

A transitive dependency is formed by two functional dependencies.

Z is transitively dependent on X if Y is functionally dependent on X but X is not functionally dependent on Y, and Z is functionally dependent on Y. In other words, X → Z is a transitive dependency if the following hold true:

- Y depends on X, X → Y
- X does not depend on Y
- Z depends on Y, Y → Z

(X)			(Y)	(Z)
MatricNo	Name	Gender	CivicsClass	CivicsTutor
1	Alex	M	22S01	Peter Tan
2	Brandon	M	22S01	Peter Tan
3	Charles	M	22S01	Peter Tan
4	Dave	M	22S01	Peter Tan
5	Eliza	F	22S05	Jane Wong
6	Felicia	F	22S11	Mary Ng
7	George	M	22S11	Mary Ng
8	Hongli	M	22S11	Mary Ng

- CivicsClass (Y) depends on MatricNo (X)
- MatricNo (X) does not depend on CivicsClass (Y)
- CivicsTutor (Z) depends on CivicsClass (Y)

Hence, CivicsTutor (Z) depends transitively on MatricNo (X)